

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)		$v = at$ and $v^2 = 2ax$ seen (i.e. consequence of $u = 0$ on equations) (1) $a^2t^2 = 2ax$ seen (1) [implies first mark]	1	1		2	2	
	(b)		Graph 3 (1) Vertical: constant acceleration due to [force of] gravity (1) Horizontal: no acceleration (accept constant speed or constant velocity) because no forces act (1)	1	1 1		3	1	
	(c)	(i)	$t = \frac{1.8}{3.4}$ [= 0.53 s] (1) Height = $\frac{1}{2} \times 9.81 \times 0.53^2$ (1) substitution and $u = 0$ (ecf on t) Height = 1.37 [m] (1)	1	1 1		3	3	
		(ii)	Vertical velocity = $9.81 \times 0.53 = 5.2$ [m s^{-1}] (1) (ecf on t) [Alternative: vertical velocity = $((2 \times 9.81 \times 1.37)^{1/2})$ (ecf on h) Pythagoras: $V_R^2 = 5.2^2 + 3.4^2$ (1) (ecf on vertical velocity) $V_R = 6.2$ [m s^{-1}] (1) At 57° to the horizontal (1) (apply ecf if incorrect vertical or resultant velocity used to calculate angle)	1	1 1 1		4	4	
	(d)		Untrue and link to ... Flight time depends on (two $\times$ (1) from): • drop height • acceleration [due to gravity] • [initial] vertical velocity [Award 1 mark only for untrue because horizontal and vertical motions are independent of each other]			2	2		
			Question 5 total	4	8	2	14	10	0